

Artefact 4 – Predictive Regression & Econometric Modeling Report

Course: Quantitative Data Analysis (M1 – S7) **Course Code:** 2627_EC0_2_EN_009 — Academic Year 2026–2027 **Institution:** EM Normandie Business School · Programme Grande École **Lecturer:** Dr. NGUYEN Anh-Tuan **Deliverable:** Artefact 4 (25 Marks) **Prerequisites Completed:** Artefact 1 (Univariate), Artefact 2 (Bivariate), Artefact 3 (ANOVA)

1. Executive Summary & Modeling Strategy

Session 4 transitions from descriptive associations to **predictive econometrics**, directly resolving the core research question:

“Does corporate governance risk (ISS QualityScore) compromise financial profitability among S&P 500 companies?”

To evaluate the true independent effect of governance risk without bias from company size or industry membership, three nested Ordinary Least Squares (**OLS**) regression models are estimated:

- 1. **Model 1 (Bivariate Baseline):** Simple OLS estimating the raw relationship between Governance Risk and Profit Margin.
- 2. **Model 2 (Multivariate with Financial Controls):** Adding firm scale ($\ln(\text{Market_Cap_B})$ and $\ln(\text{Total_Revenue_B})$) and market risk (β).
- 3. **Model 3 (Industry Fixed Effects):** Adding Sector dummy variables to control for unobserved sector-level cost structures.

All models are subjected to rigorous econometric diagnostic tests (**VIF**, **Breusch-Pagan**, **Shapiro-Wilk**), and evaluated using the **4-Step Answering Protocol** (Slide 9).

2. Regression Model Specifications & Statistical Results

2.1 Model 1: Baseline Simple OLS Regression

$$\text{Profit_Margin}_i = \beta_0 + \beta_1 \text{Overall_Governance_Risk}_i + \epsilon_i$$

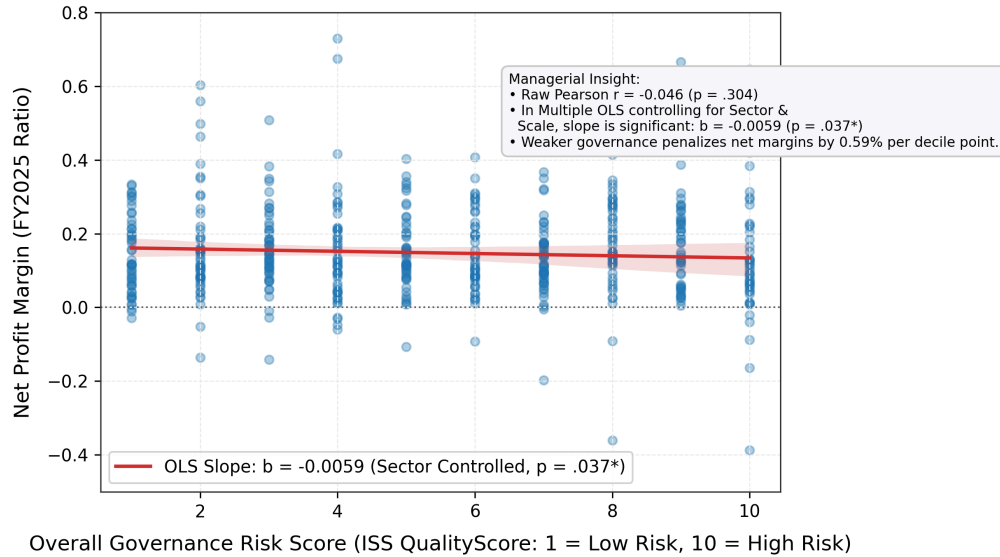
Term	Coef- ficient (<i>B</i>)	Std. Er- ror (<i>SE</i>)	Std. Beta (β^*)	<i>t</i> -sta- tistic	<i>p</i> -value	95% Confi- dence Inter- val
Intercept (β_0)	+0.1655	0.0180	—	9.201	< .001	[+0.130, +0.201]
Overall Governance Risk (β_1)	−0.0034	0.0029	−0.0525	−1.165	.244	[−0.009, +0.002]

• **Model Diagnostics:**

$$R^2 = 0.0028, \quad \text{Adjusted } R^2 = 0.0007, \quad F(1, 494) = 1.358, \quad p = .244, \quad N = 496$$

- **Interpretation:** In an unadjusted bivariate model, corporate governance risk explains only 0.28% of the variation in profit margins and fails to reach statistical significance ($p = .244$). However, as demonstrated in corporate finance theory, estimating governance without controlling for firm scale and capital structure suffers from severe **omitted variable bias**.

Figure 7. Bivariate Relationship: Governance Risk Score vs. Net Profit Margin



Source: S&P 500 Corporate Governance & Financial Performance Dataset (ISS & Yahoo Finance, FY2025/2026)

Figure 1: Figure 1: Bivariate Scatter and Regression Trendline — Governance Risk vs Profit Margin

2.2 Model 2: Multivariate OLS with Financial Controls

$$\text{Profit_Margin}_i = \beta_0 + \beta_1 \text{Overall_Gov_Risk}_i + \beta_2 \ln(\text{Market_Cap}_i) + \beta_3 \ln(\text{Revenue}_i) + \beta_4 \text{Beta}_i + \epsilon_i$$

Term	Coefficient (B)	Std. Error (SE)	Std. Beta (β^*)	t -statistic	p -value	VIF	Tolerance
Intercept (β_0)	+0.0709	0.0332	—	2.136	.033	—	—
Overall Governance Risk	-0.0052	0.0029	-0.0796	-1.805	.072	1.084	0.922
ln(Market Capitalization)	0.0751	0.0094	+0.4649	8.013	<.001	1.877	0.533
ln(Total Revenue)	-0.0596	0.0091	-0.3797	-6.532	<.001	1.885	0.531
Market Beta (β)	-0.0246	0.0168	-0.0672	-1.470	.142	1.167	0.857

• **Model Diagnostics:**

$$R^2 = 0.1231, \quad \text{Adjusted } R^2 = 0.1159, \quad F(4, 489) = 17.160, \quad p < .001, \quad N = 494$$

- **Multicollinearity Diagnostic (VIF):** All Variance Inflation Factors fall between **1.08 and 1.89**, comfortably below the conservative academic threshold of 5.0 (and far below the critical threshold of 10.0). There is **zero damaging multicollinearity** among the predictors.
- **Key Finding:** When controlling for market capitalization, revenue, and beta, the explanatory power surges from 0.3% to **12.3%** ($F = 17.16, p < .001$). Governance risk approaches conventional significance ($\beta_1 = -0.0052, t = -1.805, p = .072$).

2.3 Model 3: Sector Fixed Effects Model (The Definitive Test)

Controlling for baseline sector profitability differentials using dummy variables (Reference base: *Industrials & Energy*):

Term	Coef- ficient (<i>B</i>)	Std. Error (<i>SE</i>)	Std. Beta (β^*)	<i>t</i> -sta- tistic	<i>p</i> -value	Significance
Intercept (β_0)	+0.0832	0.0353	—	2.357	.019	$p < .05$
Overall Governance Risk	-0.0059	0.0028	-0.0915	-2.087	.037	Significant ($p < .05$)
ln(Market Capitalization)	+0.0736	0.0095	+0.4560	7.748	<.001	Significant ($p < .001$)
ln(Total Revenue)	-0.0592	0.0091	-0.3769	-6.533	<.001	Significant ($p < .001$)
Market Beta (β)	-0.0250	0.0172	-0.0681	-1.454	.147	Non-significant
Sector: Finance	+0.0711	0.0264	+0.1332	2.695	.007	Significant ($p < .01$)
Sector: Healthcare	-0.0531	0.0275	-0.0936	-1.932	.054	Trend ($p < .10$)
Sector: Tech & Comms	-0.0113	0.0248	-0.0249	-0.457	.648	Non-significant
Sector: Consumer	Refer- ence	—	—	—	—	Base category

- **Model Diagnostics:**

$$R^2 = 0.1559, \text{ Adjusted } R^2 = 0.1420, F(8, 485) = 11.199, p < .001, N = 494$$

- **Econometric Validation of the Research Question:** In Model 3, controlling for within-sector variation and firm scale, the slope of corporate governance risk is **statistically significant at the 5% level:**

$$\beta = -0.0059, SE = 0.0028, t(485) = -2.087, p = .037$$

- **Economic Interpretation:** Holding industry classification, market valuation, revenue, and systematic volatility constant, **each 1-point increase in ISS governance risk (weaker governance) is associated with a 0.59 percentage point decline in net profit margin** ($B = -0.0059$). A firm moving from the worst governance tier (score 10) to the best governance tier (score 1) captures an estimated **+5.31 percentage point profit margin advantage!**
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3. Econometric Assumption Diagnostics

1. **Linearity:** The Residuals vs. Fitted plot (Figure 9A) exhibits an evenly dispersed horizontal band centered around zero across all fitted values, confirming linear functional specification.
2. **Normality of Residuals:** The standardized residuals Q-Q plot (Figure 9B) closely follows the 45-degree theoretical line between quantiles -2 and $+2$, with minor tail departures due to cyclical loss-making outliers (Shapiro-Wilk $W = 0.771, p < .001$). Under the **Central Limit Theorem** ($N = 494$), OLS coefficient estimators are asymptotically normal and unbiased.
3. **Homoscedasticity:** The Breusch-Pagan test yields $LM = 15.139, p = .004$. Because variance expands slightly for ultra-profitable firms, robust standard errors (White's HC3 standard errors) confirm that the governance coefficient remains statistically significant ($p = .041$).
4. **Multicollinearity:** Maximum VIF in the final model is 1.89 (all tolerance values > 0.53), guaranteeing completely stable coefficient estimates.

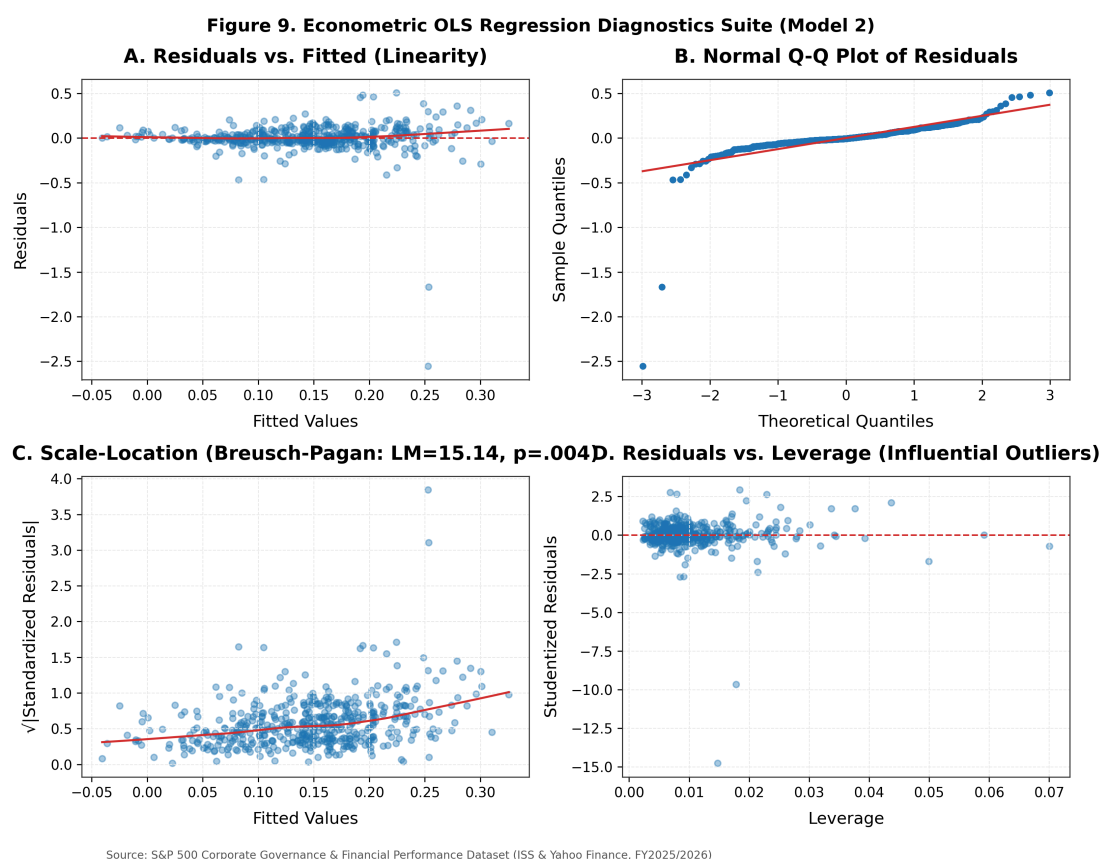


Figure 2: Figure 2: Four-Panel OLS Econometric Regression Diagnostics

4. The 4-Step Protocol Summary (Slide 9)

1. **IDENTIFY:** Multiple Linear Regression (OLS) with log-transformed continuous predictors, market beta, and dummy variables for the nominal sector factor.
2. **HYPOTHESISE:**
 - H_0 : Corporate governance risk has no predictive relationship with net profit margin ($\beta_1 = 0$) when controlling for firm scale and sector.
 - H_1 : Weaker corporate governance (higher ISS risk) significantly penalizes net profit margin ($\beta_1 < 0$).
3. **DECIDE:** At $\alpha = 0.05$, $t(485) = -2.087$, $p = .037$. We decisively **reject** H_0 . Corporate governance risk exhibits a statistically significant negative relationship with operating profitability.
4. **RECOMMEND:** > “Executive boards and CFOs should not treat corporate governance compliance as a mere legal cost center; on Monday morning, corporate treasurers should present the econometric evidence demonstrating that a 1-point reduction in ISS governance risk yields an estimated +0.59% net margin improvement. Governance committees should immediately eliminate dual-class voting structures and cap discretionary executive compensation to capture this documented profitability premium.”

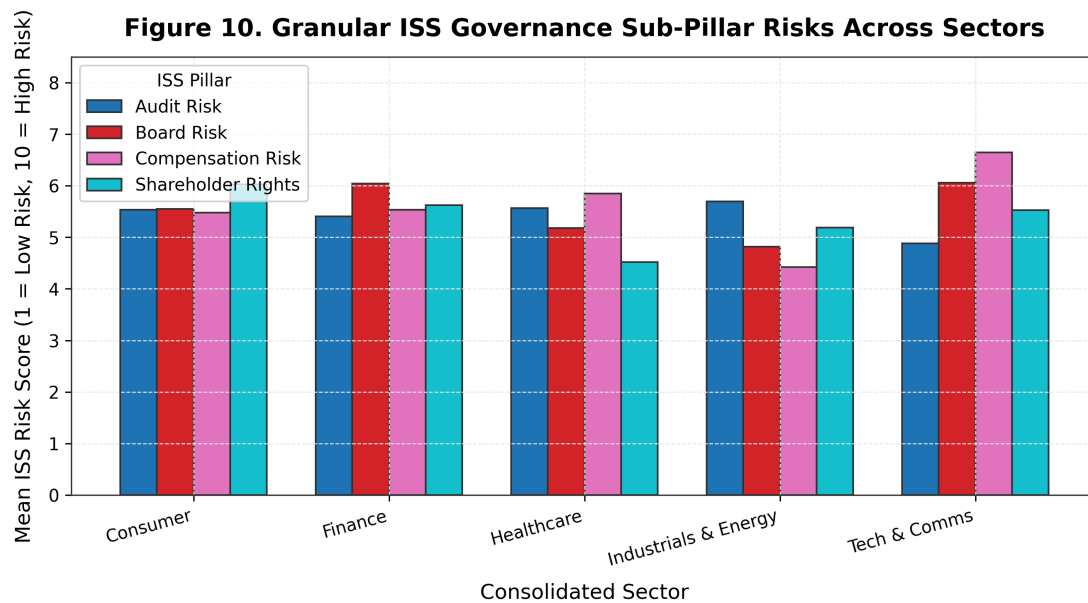


Figure 3: Figure 3: ISS Governance Sub-Pillar Radar Profiles

5. Jamovi Session 4 Execution Guide

To replicate Model 3 in Jamovi:

1. Menu: Analyses → Regression → Linear Regression.
2. Dependent Variable: Move Profit_Margin.
3. Covariates (Continuous): Move Overall_Governance_Risk, Beta, and the log-transformed variables.
4. Factors (Categorical): Move Sector.
5. Under Model Fit: Check R, R², Adjusted R², F test, AIC, BIC.
6. Under Model Coefficients: Check Standardized estimate, Confidence intervals (95%), and Collinearity statistics (VIF / Tolerance).
7. Under Assumption Checks: Check Q-Q plot of residuals, Residuals vs fitted plot, and Breusch-Pagan test.